A Weak Supervision Learning Paradigm for Oriented Ship Detection in SAR Image

Tingxuan Yue, Yanmei Zhang, Jin Wang, Yanbing Xu, and Pengyun Liu

***Abstract*— The advancement of convolutional neural networks (CNNs) has greatly promoted the development of ship detection in SAR images. Oriented ship detection is practical and imperative, but the development of its fully-supervised methods is limited by the insufficiency of data annotated in an oriented bounding box (OBB). Therefore, we proposed a weakly supervised learning (WSL) paradigm to train the network by only data annotated in a horizontal bounding box (HBB) for oriented ship detection. The paradigm follows a coarse-to-fine prediction route, and the ships' coarse orientations and high-quality pseudolabels both mined from images are used to train the network combined with the HBB annotations. The designed orientation initialization decoder (OID) utilizes the output from the prediction branch of ship coarse orientation (PBSCO) and the regression branch to decode coarse OBBs, based on the horizontal proposals, and then these OBBs are refined by the fine prediction head, which is jointly trained by high-quality pseudolabels and HBB annotation. Extensive experiments were conducted on the SAR ship detection dataset (SSDD) and the high-resolution SAR images dataset (HRSID) with both OBB and HBB annotations, and the results show that our proposed WSL paradigm can train network to attain the AP_{50} and R in the same level of mainstream fully-supervised methods in remote sensing.**

***Index Terms*— Convolutional neural network (CNN), oriented ship detection, remote sensing, synthetic aperture radar (SAR), weekly supervised learning (WSL).**

I. Introduction

SYNTHETIC aperture radar (SAR) is an all-weather and all-day imaging system in remote sensing [1]. Due to the demand for ship monitoring in military and civilian, ship detectors in SAR image have developed rapidly [2], [3]. With the popularity of deep learning technology, the ship detection methods based on convolutional neural networks (CNNs) have shown remarkable performance [4], [5].

Many common object detection methods based on CNN have been migrated to SAR ship detection. These methods can be divided into the anchor-based and the anchor-free, and the anchor-based methods can be divided into the single-stage and the multistage. The RetinaNet [6] and faster R-CNN [7] are the basal network structures of the single-stage and the multistage in anchor-based methods, and the FCOS [8] is the common network structure of the anchor-free methods. There are several challenges in SAR ship detection, such as the multiscale of the ships, the false alarms caused by islands and buildings and the interference from adjacent ships and coasts. The first challenge is usually treated by the scale adaptation of anchors and feature maps [9]. Wei et al. [11] introduced the low-level feature map to improve the prediction scale coverage of FPN [10] based on the structure of faster R-CNN. The anchors generated by the network can effectively capture the multiscale ships in SAR image [12]. The multiscale pooling operation can also keep the information of multiscale ships in the feature maps, especially small ships [13]. The last two challenges can be addressed by enhancing the representational capacity of feature maps [14]. Hu et al. [15] applied the deformable convolution and nonlocal attention mechanism on the feature map to modify the FCOS for SAR ship detection. Fu et al. [16] proposed a module and inserted it into the FCOS for mitigating interference from adjacent objects. Sun et al. [17] fused category information and position information from feature maps to improve the performance of the FCOS network in detecting ships. In addition, some structures designed specifically for the SAR sensor can enrich the semantic information of feature maps, which are able to improve the performance of ship detection in SAR images effectively [18], [19], [20], [21].

Compared with common object detection, the horizontal bounding box (HBB) can not fully meet the requirements for object detection in remote sensing. The trajectory prediction of ships and the formation analysis of fleets both require the bounding boxes containing the angle of ships. Therefore, the oriented bounding box (OBB) detection is an important research hotspot [22], and the OBB can provide purer ship chips for the ship classification tasks because it fits the ship more closely. Some innovative loss functions, such as CSL [23], GWD [24], and KFIoU [25], were proposed to solve the problems of angle boundary discontinuity in the OBB prediction. The task-collaborated OBB regression and classification can improve the performance of oriented object detectors [26]. In addition, the OBB encoding representation and network structures were redesigned specifically for the OBB detection [27], [28], [29]. These methods are transferred to SAR ship detection from optical remote sensing through adaptive modification for dealing with the three challenges. A novel OBB encoding representation was proposed to break the border restriction of the HBB [30], which is more suitable for SAR ship detection compared with the midpoint offsets representation in [27]. Some branches and modules designed

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TABLE I
MAINSTREAM SAR SHIP DETECTION DATASETS

Year	Datasets	Annotation
2017	SSDD [38]	Horizontal
2019	SAR-Ship-Dataset [39]	Horizontal
2019	AIR-SARShip-1.0 [40]	Horizontal
2019	AIR-SARShip-2.0 [40]	Horizontal
2020	HRSID [41]	Horizontal & Oriented
2020	LS-SSDD-v1.0 [42]	Horizontal
2020	Official-SSDD [43]	Horizontal & Oriented
2021	SRSDD-v1.0 [44]	Oriented
2022	RSDD-SAR [45]	Oriented

specifically were added in the mainstream networks for the SAR ship OBB detection, and they focused on angle prediction and generation of more compatible feature [31], [32], [33]. Additionally, the scattering points are characteristic of objects in the SAR image [34], and these points can be used to predict OBB [35], [36], [37].

These above methods are all fully-supervised detection methods, and their performance depends heavily on the training dataset. Since annotation of ships in SAR imagery usually requires the assistance of sensors in other domains, producing the SAR ship detection datasets needs more effort compared with optical remote sensing imagery. Enclosing ships tightly to obtain an OBB annotation is more cumbersome than the click-and-drag to obtain an HBB annotation in labeling software. Therefore, the released SAR ship detection datasets are few in number and mainly annotated as HBB as shown in Table I. Although OBB detection is more practical, the lack of datasets limits the further development of oriented SAR ship detection. The price of OBB annotation is approximately 1.4 times that of HBB annotation in the Google AI Platform.[1] To avoid the expensive cost of OBB annotation, the weakly supervised learning (WSL) methods are investigated for oriented object detection in training using only HBB annotation.

In this article, we proposed a WSL paradigm for oriented ship detection in SAR image, only trained by HBB annotation SAR ship data. Specifically, this WSL paradigm is a self-supervised pattern using ship characteristics in the SAR images, and it can serve as a basis for developing other WSL ship detection methods in SAR images. Our paradigm combines ships' coarse orientations and high-quality pseudolabels to guide the learning of prediction accurate OBBs under a geometric projection constraint. The geometric projection constraint is shown as (a) of Fig. 1, and the projection HBB projected by the prediction OBB is used to calculate the loss with the HBB annotation in the network training as shown in (b) of Fig. 1. The pseudolabels are generated directly from SAR images without additional conversion processing due to the fact that SAR images are grayscale images. First, each ship in the horizontal chips is smoother and more complete after image morphology processing, and pseudolabels are the minimum area rectangles of each ship in the chips. Then the high-quality pseudolabels are selected by a set range of long edge ratio. There are not sufficient high-quality pseudolabels

[1]https://cloud.google.com/ai-platform/data-labeling/pricing

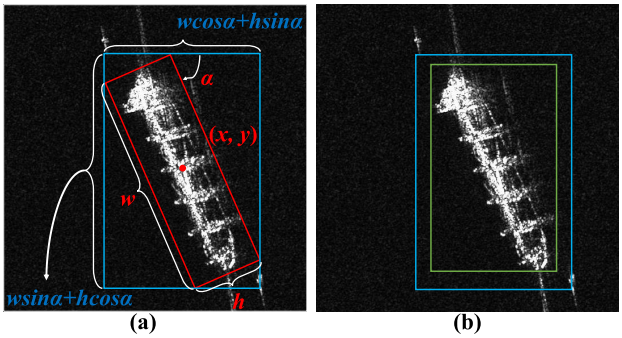


Fig. 1. Comparison of different bounding boxes. (a) Geometric projection constraint, and the blue and red bounding boxes are projection HBB and prediction OBB, respectively. The (x, y, w, h, α) are five parameters of the prediction OBB. (b) Gap between projection HBB and HBB annotation, and the blue and green bounding boxes are projection HBB and HBB annotation, respectively.

to train the network to the performance of full supervision caused by the interference from adjacent coasts and ships, so we followed a coarse-to-fine prediction route, and the high-quality pseudolabels are applied in the stage of fine prediction. In the stage of coarse prediction, the ships' coarse orientations are extracted from the horizontal ship chips and used to train the prediction branch of ship coarse orientation (PBSCO). And the coarse prediction OBBs are decoded by the orientation initialization decoder (OID) through the three inputs: orientations output from PBSCO, angle offsets output from the angle branch and coded values output from the regression branch. Since there is a gap between annotated HBB and projected HBB as shown in (b) of Fig. 1 and there are only SSDD and HRSID datasets that have both standard annotation of HBB and OBB as shown in Table I, we conducted abundant experiments on the two datasets to verify the effectiveness of our paradigm. The major contributions of our work can be summarized as follows.

1) To our knowledge, the paradigm we proposed is the first WSL paradigm specifically designed for SAR-oriented ship detection, which requires only the training of HBB annotation. And the paradigm follows a coarse-to-fine route for predicting OBBs directly.
2) In the coarse prediction stage, the PBSCO and OID are proposed to assist the geometric projection to constrain the learning of angle prediction. The OID utilizes orientations output from PBSCO to set an initial angle for reducing the complexity of angle regression.
3) In the fine prediction stage, the high-quality pseudolabels are generated and selected from the SAR image to train the network for predicting the fine OBBs.
4) We conducted extensive experiments on the SSDD and HRSID datasets, and the results show that the proposed paradigm can train the network to achieve the performance of the fully-supervised methods.

The remainder of this paper is organized as follows: In the Section II, the related works are summarized. The Section III presents the details of our paradigm. The Section IV analyzes the effectiveness of our paradigm with experimental data, and finally, a conclusion is drawn in Section V.

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II. Related Works

The fully supervised OBB ship detection methods in SAR images have attracted the main attention of scholars, and under the supervision training of the OBB annotations, these methods have the advantages of higher performance than the methods requiring manual design features [46], [47]. Our WSL paradigm aims to obtain an OBB ship detector under only the supervision training of the HBB annotations, which is the significant difference between our approach and existing OBB ship detection methods in SAR images. In addition, the meaning of paradigm in this article is a pattern used to train oriented ship detection by WSL in SAR images. The WSL means inexact supervision, more precisely, the ability of the network to predict the angle and shape is not trained on annotation supervision by the training data.

Zhou et al. [48] proposed a WSL method of OBB ship detector based on the transfer learning [48], but the method is an incomplete supervision using partial OBB annotation data in the training stage. The existing WSL paradigm from HBB training to predicted OBB in object detection can be summarized into two types.

A. HBB Training to Predicted OBB by Transfer Learning

The training flow of this paradigm type is training the network by the guide of the teacher network. The transfer learning is a typical framework of WSL, in which the teacher network with generalization ability guides the student network training [48], and the teacher network is usually trained by source domain data such as OBB annotation dataset in other domains. Zhu et al. [49] proposed a method based on this paradigm named KCR to obtain an OBB prediction network by sharing network parameters between the teacher network and the student network. In this method, the teacher network needs to be trained by a large source domain dataset, such as COCO [50] and DOTA [51]. In addition, indirect transfer learning is to train the student network by the pseudolabels generated from the teacher network. Hua et al. [52] trained the student network by the consistency constraint between the sets of pseudolabels and predictions, and this method requires that the teacher network can generate high-quality pseudolabels from target domain data. Training a high-performance teacher network is inefficient and costly, and there is a cross-domain obstacle from the multichannel source data to the single-channel SAR target data.

B. HBB Training to Predicted OBB by Self-Supervision

The self-supervision means that the network is trained by the network itself and HBB annotation, without additional annotation or auxiliary means. Sun et al. [53] mined the angle of the oriented target by rotating the images, then they used the angle of targets to train the network in combination with HBB annotation. The self-supervision training also contains the training by the pseudolabels generated from the network itself, and the WSODet [54] used the heat map based on feature map to generate high-quality pseudolabels for training the prediction OBBs branch, but this pseudolabels extracting method is not suitable for the SAR image due to the imaging mechanism of SAR image [55]. Yang et al. [56] proposed an innovative paradigm named H2RBox, and the H2RBox consists of geometric projection constraint and multiview consistency constraint. There are plenty of wrong prediction OBBs meeting the constraint as shown in Fig. 2, and the H2RBox utilizes both shape (w, h) invariance and angle(α) relationship of targets in the images under different views, to eliminate the wrong prediction OBBs. For solving the problem of relying on large amounts of training data in H2RBox, the flip consistency and rotate consistency of symmetrical targets have been used in H2RBoxV2 [57]. Using the two consistencies allows the network to learn prediction angle directly rather than just through geometric projection. Unfortunately, some ships in the SAR image lost their symmetry caused by structural incompleteness and scattered noise both brought by the SAR imaging process, and the amount of SAR ship data which is available for the training is inadequate compared with optical remote sensing data.

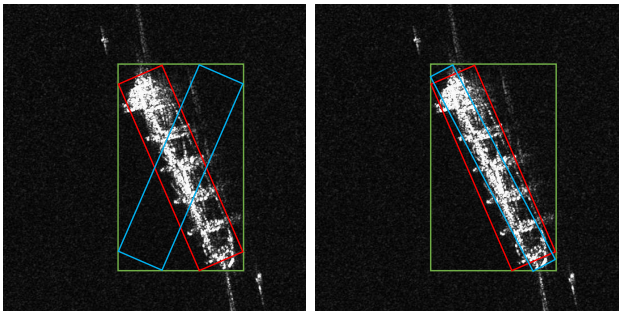


Fig. 2. Two typical wrong prediction OBBs under geometric projection constraint only, and the green, blue, and red bounding boxes are the annotation HBB, wrong prediction OBB, and the accurate prediction OBB, respectively. (a) Prediction OBB of the wrong angle. (b) Prediction OBB of the wrong shape.

III. Proposed Method

This section describes the proposed paradigm as shown in Fig. 3 in detail. The multistage network is the basal structure of the proposed paradigm, and the WSL paradigm for SAR ship detection is trained by HBB annotation and information mined from the SAR image: orientations and pseudolabels. The long edge definition 90° (Le90) is the definition of an OBB in the proposed paradigm, and the (x, y, w, h, α) are five parameters of an OBB in the Le90, where (x, y), (w, h) and α are center localization, shape, and angle of long edge respectively, and the interval of α is $[-90°, 90°)$. There are coarse prediction and fine prediction in turn in the paradigm under the coarse-to-fine route as shown in Fig. 3, and the combination of two prediction stages can eliminate wrong prediction OBBs as shown in Fig. 2. In the coarse prediction, the coarse OBBs are decoded by the OID based on the horizontal proposals. The outputs of PBSCO, regression branch, and angle branch are fed into the OID, then the predicted OBBs with the wrong angle as shown in (a) of Fig. 2 are regulated by the orientations and angles in the OID. The three branches are trained by the GT annotation and the output of the coarse orientation extractor. Subsequently, these coarse OBBs are the decoded

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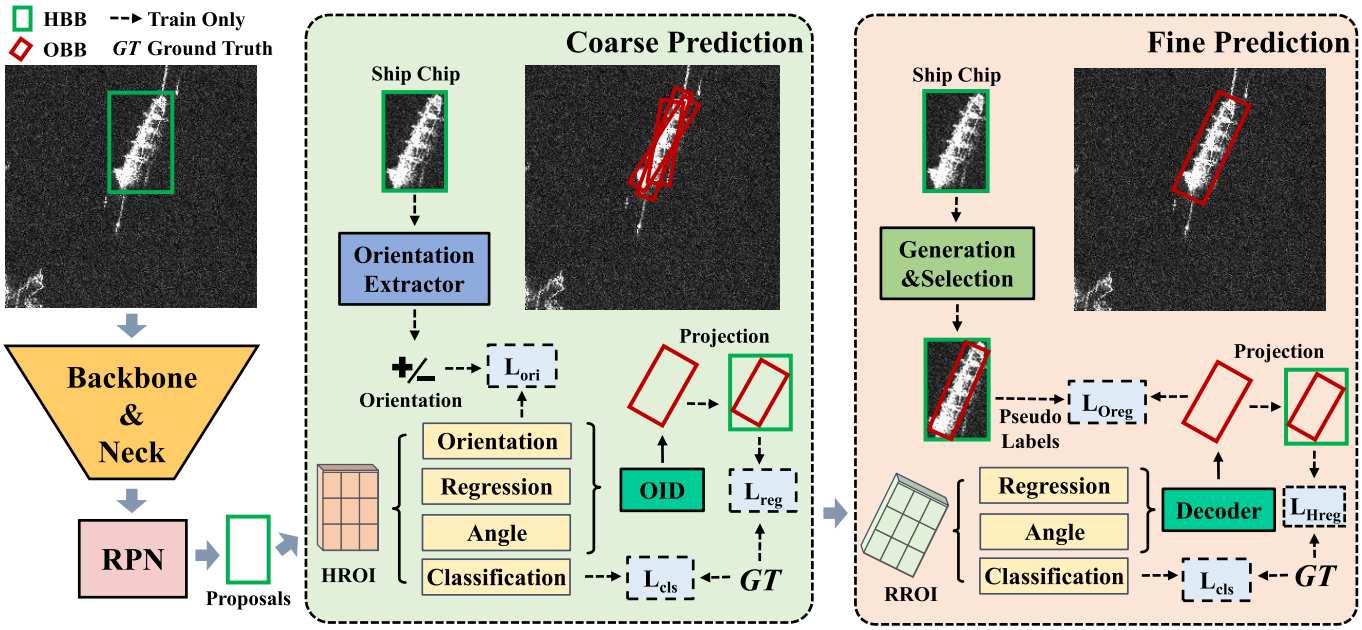


Fig. 3. Overall framework of the paradigm, and the GT is annotated by HBB. The HROI and RROI are the abbreviations of horizontal and rotated regions of interest. The '+' and '−' of orientation are the symbols of the angle value in the Le90 representation. The L_{ori}, L_{cls}, and L_{reg} stand for the loss functions of orientation, classification, and regression, respectively. The L_{Oreg} and L_{Hreg} in the Fine Prediction stand for the loss functions of OBB regression and HBB regression, respectively.

bases to obtain the fine OBBs. The high-quality pseudolabels are generated and selected from the ship chips for training the fine prediction head with the GT annotation, and the network can predict the accurate shapes of OBBs when the training is done.

A. Network Framework

Our WSL-oriented ships detection network is similar in structure to the common network training by the OBB annotation, with the only difference of PBSCO. The backbone, neck, and RPN in Fig. 3 are the same as the Faster RCNN using in HBB prediction, i.e., the ResNet50, the FPN and the RPN outputting horizontal proposals, and the loss of RPN is calculated by horizontal samples and the GT annotated in HBB.

The horizontal proposals are as bases to predict the OBBs by the coarse and fine prediction heads as shown in Fig. 3. The feature of HROI is extracted by the RoI Pooling and then fed into the two sharing FCs, and there are four individual FCs for the four branches after the two sharing FCs, as shown in (a) of Fig. 4. The coarse prediction OBBs are used to obtain rotated features by RROI Pooling from the corresponding feature map P output from FPN, and each RROI is divided into 7×7 grids, and the feature of each gird can be calculated by

$$\sum_{(x,y)\in gird} P(T(x, y, \alpha))/n \tag{1}$$

where $T(\cdot)$ is rotated transformation from [28], and n is the number of sample points in the gird. These features of RROIs are fed into the fine prediction head, and there are three individual FCs as the three branches, like the branches of the coarse prediction head. A convolution using 3×3 kernel is inserted before the two sharing FCs for the interaction within the feature as shown in (b) of Fig. 4. When calculating the loss with GT, the oriented samples are projected to the horizontal (x_1, y_1, x_2, y_2) as shown in (a) of Fig. 1, and the projection is calculated as follows:

$$\text{Projection}(x, y, w, h, \alpha) = \begin{cases} x_1 = x - \frac{1}{2}(w\cos\alpha + h\sin\alpha) \\ y_1 = y - \frac{1}{2}(w\sin\alpha + h\cos\alpha) \\ x_2 = x + \frac{1}{2}(w\cos\alpha + h\sin\alpha) \\ y_2 = y + \frac{1}{2}(w\sin\alpha + h\cos\alpha). \end{cases} \tag{2}$$

B. PBSCO and OID

The PBSCO is shown as (a) of Fig. 4, and it is parallel to other prediction branches followed by the two sharing FC. The coarse means that the orientations used to train the branch are coarse because the extracted orientations may be reversed in rare cases. The coarse orientations for the PBSCO are mined from the ship chips extracted in the SAR image, and the typical chips are shown as Fig. 5. The flow of the coarse orientation extractor is shown as the Algorithm 1, and the input of the coarse orientation extractor is a ship chip. The binary image $\mathbf{I}_b$ can be obtained as Lines 1 to 2, and the major ship structure is preserved in $\mathbf{I}_b$. The Mean() in Line 1 stands for gaining the average gray value of the input image. As Lines 3 to 4, the gray values summation of four parts S_{tl}, S_{tr}, S_{bl}, and S_{br} are calculated, where S_{tl}, S_{tr}, S_{bl}, and S_{br} correspond to four

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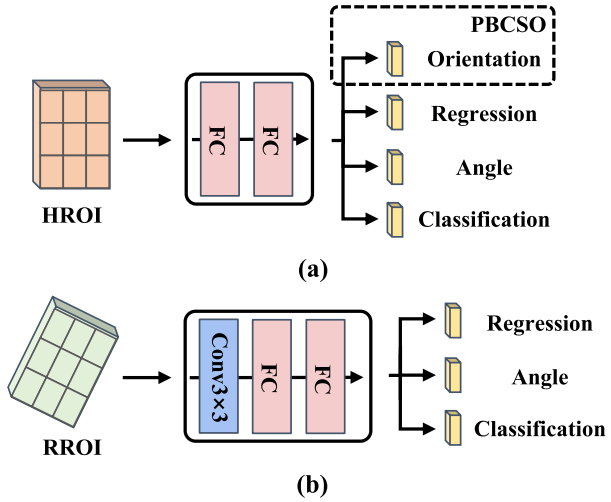


Fig. 4. Network structures of the coarse and fine prediction head. The FC and Conv3 × 3 are abbreviations of fully connected layer and convolution layer using 3 × 3 kernel respectively. (a) Network structure of the coarse prediction head. (b) Network structure of the fine prediction head.

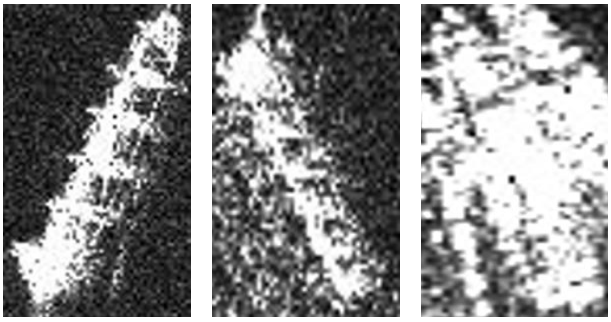


Fig. 5. Typical ship chips extracted from the SAR image. There are individual ships, shoreside ships, and adjacent ships, respectively.

parts of top left, top right, bottom left and bottom right in the chip, respectively, as shown in Fig. 6. The orientation **O** can be obtained according to Lines 5 to 10, and the Min() in Line 5 stands for gaining the minimum. The **O** indicates the symbol of ship angle using the Le90, and the **O** of the three ships in Fig. 6 calculated by Algorithm 1 are '−,' '+,' and '+' in accord with the facts of Fig. 5. The binary classification prediction in the PBSCO is trained by the **O**, and the **O** is transformed to '−1' or '+1' for regulating the angle θ of coarse prediction OBB as follows:

$$\theta_r = \mathbf{O} * |\theta| \tag{3}$$

where θ_r and $|\theta|$ are regulated θ and gained absolute value of θ, respectively.

To reduce the complexity of angle regression, the OID is proposed to utilize the **O** and the characteristics of HBB. The OID is proposed based on a phenomenon that one of the diagonal angles of its HBB is similar to the angle of its OBB for an elongated ship in most cases as shown in Fig. 5. Even for horizontal and vertical ships, using the diagonal angle of HBB is simpler and more stable than using a random angle, as the initial angle of OBB in the angle regression. The predicted values $(x_t, y_t, w_t, h_t, \theta_t)$ from the coarse prediction stage are fed into the OID. And the parameters (x, y, w, h, θ)

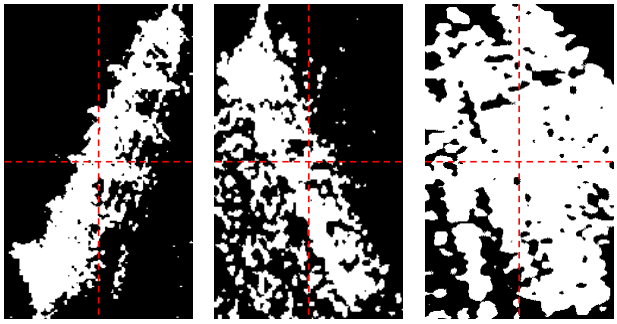


Fig. 6. Three typical ship chips divided into four parts. The focus of the horizontal and vertical red dashed lines is the center point of the chip.

Algorithm 1 Flow of Coarse Orientation Extractor

Input:
The ship chip extracted from the SAR image **I**.
Output:
The orientation of the ship in the chip **O**.
1: To calculate the average gray value V_{ag}: $V_{ag} = Mean(\mathbf{I})$;
2: To obtain a binary image $\mathbf{I}_b$ by threshold V_{ag};
3: To divide the $\mathbf{I}_b$ into four parts as shown in the Fig. 6;
4: To sum the gray values of four parts S_{tl}, S_{tr}, S_{bl} and S_{br};
5: To obtain the representative summation S_+ and S_-:
$S_+ = Min(S_{tl}, S_{br})$, $S_- = Min(S_{tr}, S_{bl})$;
6: **if** $S_+ > S_-$ **then**
7: $\mathbf{O} \leftarrow +$;
8: **else**
9: $\mathbf{O} \leftarrow -$;
10: **end if**
11: Return **O**.

of OBB are decoded from the predicted values and the parameters of the horizontal proposal (x_p, y_p, w_p, h_p), where (x_p, y_p) and (w_p, h_p) are the center localization and shape of the proposal, respectively. The decoding method of (x, y, w, h) in the OID is the same as the Faster RCNN as follows:

$$x = x_t w_a + w_a, \quad y = y_t h_a + h_a \tag{4}$$

$$w = w_a e^{w_t}, \quad h = h_a e^{h_a}. \tag{5}$$

The e in (5) is the base of the natural logarithmic function. For the θ, it is decoded by the θ_t and orientation in the OID. The angle θ_{dl} of diagonal line can be calculated by (6), where arctan() is the calculation of the arc tangent

$$\theta_{\text{dl}} = \pm \arctan(h_p / w_p). \tag{6}$$

The init angle θ_{init} can be obtained as follows:

$$\theta_{\text{init}} = \mathbf{O} * |\theta_{\text{dl}}|. \tag{7}$$

And the θ of OBB is calculated as follows:

$$\theta = \theta_{\text{init}} + \theta_t. \tag{8}$$

C. Acquisition of High-Quality Pseudolabels

The high-quality pseudolabels are used to train the fine prediction head as shown in Fig. 3, and the high-quality means that the pseudolabels bind the ship's major structure, mainly

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reserving the completion of the ship's longitudinal structure. There is almost no need to consider the width of the ship when annotating the HBB, so it is difficult to set criteria about width to select the pseudolabels for multiwidth ships. Fortunately, this problem can be effectively solved by using the GT annotated in HBB to constraint the projected OBB. Therefore, the fine prediction head is trained by the combination of high-quality pseudolabels and the GT annotated in HBB.

The high-quality pseudolabels are generated and selected as shown in the Algorithm 2. The noise in the chip is filtered out by the mean filtering in Line 1, and the tiny interstices of ship structure are padded through the 3 × 3 kernel. In Line 2, the close$_{x\times x}$() is the morphological closing operation for an input image, where $x \times x$ is the size of the operation kernel. The gap in the ship caused by imaging mechanisms can be effectively eliminated by the morphological closing operation, and the size of the operation kernel is selected according to the size of a chip. The Max() in Line 2 is gaining the maximum value of the input. Then the chip is binarized by the OSTU [58] for generating the minimum area rectangle, and another closing operation is used to eliminate gaps in the binarized ship structure in Line 4. The minAreaRect in Line 5 is to obtain a minimum area rectangle for ship structure, i.e., the largest contiguous area in the $\mathbf{I}_b'''$. In the end, the high-quality pseudolabel is selected by the ratio of the pseudolabel's long edge and the chip's diagonal line, like Lines 6 to 11. The range of ratio is the empirical value obtained through many experiments. Since the Le90 is used in the proposed paradigm, the high-quality pseudolabel is transformed to the format of Le90 in Line 9.

Algorithm 2 Flow of Generating and Selecting High-Quality Pseudolabels

Input:
The ship chip extracted from the SAR image $\mathbf{I}$.

Output:
The parameters of the high-quality pseudo-label for the ship in the chip $(x_{pl}, y_{pl}, w_{pl}, h_{pl}, \alpha_{pl})$.

1: To obtain a blur image $\mathbf{I}_b$ by mean filtering:
$\mathbf{I}_b = blur_{3\times3}(\mathbf{I})$;
2: To apply morphological closing on $\mathbf{I}_b$ to obtain $\mathbf{I}_b'$:
if $Max(w, h) < 60$ then
$\mathbf{I}_b' = close_{3\times3}(\mathbf{I}_b)$;
else
$\mathbf{I}_b' = close_{3\times3}(close_{5\times5}(\mathbf{I}_b))$;
3: To obtain a binary image $\mathbf{I}_b''$ by the OTSU;
4: To apply morphological closing on $\mathbf{I}_b''$ to obtain $\mathbf{I}_b'''$:
$\mathbf{I}_b''' = close_{3\times3}(\mathbf{I}_b'')$;
5: To generate an OBB for bounding the largest contiguous area in $\mathbf{I}_b'''$:
OBB $\leftarrow minAreaRect(\mathbf{I}_b''')$;
6: To gain the length of long edge in OBB L;
7: To calculate the length of diagonal line L_{dl} of the chip by the chip's shape (w, h): L_{dl}=$\sqrt{w^2 + h^2}$;
8: **if** $0.8 < L/L_{dl} < 1.1$ **then**
9: To transform the OBB as the Le90:
$(x_{pl}, y_{pl}, w_{pl}, h_{pl}, \alpha_{pl})$=$Le90$(OBB);
10: Return $(x_{pl}, y_{pl}, w_{pl}, h_{pl}, \alpha_{pl})$.
11: **end if**

D. Others

There are other details in Fig. 3 at this subsection, including classification branches, loss functions, samples assigner, and hyperparameter setting.

1) Classification Branches: The branches of classification also follow the coarse-to-fine route like the branches of prediction OBB as shown in Fig. 4. In the coarse prediction, the features of HROIs are used to predict the coarse OBBs, and then the features in the fine prediction are extracted according to RROIs. In the prediction stage, the scores used to ship classification are the average of scores from the coarse and fine prediction.

2) Loss Functions: Here is the introduction of loss functions in Fig. 3, and the q and p in each loss function are the sample and its GT in a mini-batch. The widely used Cross Entropy loss is used as L_{ori} in the PBSCO and L_{cls} in two classification branches, and the Cross Entropy loss is calculated as follows:

$$\text{Cross Entropy loss}(p, q) = -\sum_{i=1}^{C} p_i \log(q_i) \tag{9}$$

where C is the number of classifications, and the C is 2 in the orientation and the classification in the SAR ship detection. The IoU loss is the L_{reg} in the coarse prediction, and the IoU loss is calculated as follows:

$$\text{IoU loss} = -\ln \frac{\text{Intersection}(p, q)}{\text{Union}(p, q)}. \tag{10}$$

The smooth L1 loss is as L_{Oreg} and L_{Hreg} in the fine prediction, and the smooth L1 is calculated as follows:

$$\text{smoothL1 loss}(p, q) = \begin{cases} 0.5(q - p)^2, & \text{if } |x| < 1 \\ |q - p| - 0.5, & \text{otherwise.} \end{cases} \tag{11}$$

In our paradigm, the loss function of coarse prediction (L_{coarse}) for a sample is calculated as follows:

$$L_{\text{coarse}} = L_{\text{ori}} + L_{\text{cls}} + L_{\text{reg}} \tag{12}$$

and the loss function of fine prediction (L_{fine}) for a sample is calculated as follows:

$$L_{\text{fine}} = \lambda L_{\text{Oreg}} + L_{\text{cls}} + L_{\text{Hreg}}. \tag{13}$$

3) Samples Assigner: The classical MaxIoU assigner is the assignment method for positive and negative samples in our paradigm. The sample assignment results of the coarse prediction are applied in the training of PBSCO. The samples for calculating the L_{Oreg} with pseudolabels are assigned by MaxIoU assigner after the samples and pseudolabels are projected as HBB.

4) Hyperparameter Setting: There is only one hyperparameter λ of (13) in our proposed paradigm, and the λ is set by 0.25 to balance the L_{Oreg} with other two items in the L_{fine}.

IV. Experiments

To verify the effectiveness of our proposed paradigm, we performed extensive experiments on a computer with

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TABLE II
CONFIGURATION OF OPTIMIZER

Parameter	Value
Type	SGD
Initial learning rate	0.005
Epochs of learning rate decay	[11, 14]
Decay rate of learning rate	0.1
Momentum	0.9
Weight decay	0.0001

TABLE III
RESULT OF ABLATION EXPERIMENTS ON SSDD(%)

PBSCO	OID	HL	R	AP_{50}
×	×	×	72.6	58.9
✓	×	×	86.8(+14.2)	82.1(+23.2)
×	×	✓	78.9(+6.3)	73.3(+14.4)
✓	✓	×	88.5(+15.9)	85.0(+26.1)
✓	×	✓	88.7(+16.1)	86.4(+27.5)
✓	✓	✓	**89.7**(+17.1)	**87.3**(+28.4)

NVIDIA RTX 3090 GPU. Our paradigm and compared fully supervised methods are developed based on MMRotate [59], which is a rotated object detection toolbox based on the Pytorch framework.

A. Datasets and Settings

There are many open-source SAR datasets made by images produced from the satellites carrying SAR as shown in Table I, and these satellites have provided images with a resolution coverage of 0.5 to 15 m, such as TanDEM, RadarSat-2, TerraSAR-X, and Sentinel-1. For the WSL paradigm using HBB annotation to train the network with the ability to predict OBBs, there is a significant difference between the HBB annotation and the HBB projected by the OBB annotation because of their gap as shown in (b) of Fig. 1. Therefore, only the two SAR ship datasets in Table I, SSDD and HRSID can be used to verify the effectiveness of our proposed paradigm, because only the two multisource and multiresolution datasets are annotated in both HBB and OBB.

There are 2456 ships in 1160 images of SSDD, and the side length of images is between 500 and 600 pixels. The image size of the HRSID is 800 × 800 pixels, and the HRSID has a total of 5604 images and 16 951 ships in different sizes. The SSDD is divided into the 232 images of the training set and the 928 images of the testing set according to [38], and the ratio of image assignment of HRSID is 65%: 35% according to the [41], i.e., there are 3642 images and 1962 images in the training set and the testing set, respectively. In addition, the input sizes of images in SSDD and HRSID are 512 × 512 pixels and 800 × 800 pixels, respectively, consistent with the sizes adopted in other methods. The aspect ratios of input images are kept, and resizing images are using zero-padding operation.

There is no data augmentation method used before the training of networks to verify the effectiveness of the paradigm objectively. These networks are loaded with the pretrained ResNet50 checkpoint trained on ImageNet, except the ReDet, and the ReDet is a loaded checkpoint of ReResNet trained on DOTA at the beginning of training. All networks are trained on GPU with a batchsize of 2 for 15 epochs, and the configurations of the optimizer are shown in Table II. A linear warm-up strategy is adopted in the early stage of training, and its warm-up iterations and the warm-up rate are setting by 500 and 1/3, respectively. The rest of training and testing parameters follow the default settings of the Faster RCNN in the MMRotate.

B. Evaluation Metrics

Some conventional and quantitative metrics are adopted to evaluate the performance of these methods, such as the most widely used recall (R) and precision (P), and they are defined by

$$R = \frac{N_{\mathrm{TP}}}{N_{\mathrm{TP}} + N_{\mathrm{FN}}} \tag{14}$$

$$P = \frac{N_{\mathrm{TP}}}{N_{\mathrm{TP}} + N_{\mathrm{FP}}} \tag{15}$$

where N_{TP}, N_{FN}, and N_{FP} are numbers of true positives (TP), false positives (FP), and false negatives (FN), respectively. The average precision with IoU threshold 0.5, AP_{50}, from the evaluation metrics of PASCAL VOC [60] is introduced as a metric, and it is used to replace single P. In SAR-oriented ship detection, the TP is the correctly detected ship, with the rotated IoU >= 0.5 between the predicted OBB and GT OBB. The FP is a false alarm, or where the rotated IoU is less than 0.5 between the predicted OBB and GT OBB, and the FN stands for missed ships. In addition, the AP_{50} and R of both inshore and offshore testing ships are used to compare with the fully supervised detection methods.

C. Ablation Experiments

To verify and analyze the effectiveness of the PBSCO, the OID, and training by high-quality pseudolabels in the proposed paradigm individually and jointly, the ablation experiments are conducted in this section. The result of ablation experiments on SSDD is shown in Table III, and the PL stands for the training by the high-quality pseudolabels. The baseline without the PBSCO, OID, or HL follows the coarse-to-fine route, and only the geometric projection consistency is used in the two prediction heads. There is no experiment of a single OID because the OID can only be used by the orientation output from the PBSCO.

The AP_{50} and R of baseline are 58.9% and 72.6% in Table III, because only an OBB can be predicted but its angle and shape are inaccurate as shown in Fig. 2. The using of PBSCO improves the AP_{50} and R to 82.1% and 86.8%, meaning that some angles of predicted OBBs are regulated to the accurate range, and the deduction can be verified in Fig. 7. The angle of the OBB predicted by baseline is positive but the GT is negative, and the predicted angle becomes negative after applying the PBSCO. Even if the orientations of the OBBs can be predicted, it is complicated to regress an accurate angle from a random initial value. The complexity of angle prediction can be reduced by the combination of the PBSCO and the OID as shown in Table III, and the AP_{50} and R are improved

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TABLE IV
COMPARISON WITH THE TYPICAL DETECTION METHODS ON SSDD(%)

	Methods	Entire		Inshore		Offshore		FPS(img/s)	Parameters(M)
		R	AP_{50}	R	AP_{50}	R	AP_{50}		
Fully-supervised	R-RetinaNet [6]	84.0	79.9	67.9	59.2	89.3	86.5	22.3	36.13
	GWD [24]	85.9	82.5	65.6	55.6	92.5	90.5	22.1	36.13
	R-ATSS [61]	89.1	86.4	71.8	61.6	94.8	93.3	22.5	36.02
	S2ANet [62]	90.0	88.4	76.3	70.7	94.5	93.1	17.9	36.21
	R-RTMDet [63]	88.2	85.9	69.5	65.1	94.3	92.5	**35.7**	52.26
	R-Faster-RCNN [7]	89.8	87.7	77.1	72.7	94.0	93.5	27.9	41.12
	Oriented-RCNN [27]	89.1	87.6	77.1	73.3	93.0	92.0	23.2	41.13
	R3Det [64]	86.7	84.2	66.4	59.2	93.3	91.9	16.5	41.58
	ReDet [29]	88.7	86.8	74.8	70.0	93.3	92.3	17.8	**31.54**
	RoI Transformer [28]	**91.0**	**89.5**	**78.6**	**76.6**	**95.0**	**94.6**	23.8	55.03
Weakly-supervised	Our Paradigm	89.7	87.3	73.3	67.5	**95.0**	93.9	19.7	55.84

to 85.0% and 88.5%. The angles of OBBs are predicted more accurate as shown in Fig. 7. The predicted angle is further close to the GT after the combination use of PBSCO and OID, because the OID sets an initial angle value close to GT, thus reducing the the complexity of angle regression. Although the accurate angles of OBBs can be predicted based on the two components, other parameters, (w, h), cannot be accurately regressed as shown in Fig. 8. The AP_{50} and R are 73.3% and 78.9% with a small improvement of 14.4% and 6.3% when only applying HL, caused by the insufficient number of high-quality pseudolabels. Fortunately, the obstacle can be overcome by using other components. The combination of PBSCO and training by high-quality pseudolabels makes the AP_{50} and R up to 86.4% and 88.7%, and then the AP_{50} and R are 87.3% and 89.7% when used the OID complementally. It shows that the training by high-quality pseudolabels can equip the network with the ability to predict accurate shape of OBBs, on the basis of using other components, and this conclusion can be illustrated by Fig. 8. The predicted OBBs in the middle of Fig. 8, without training by high-quality pseudolabels, are not close to ships enough. This defect is repaired, i.e., the predicted OBBs are accurately bounding these ships, after training by the high-quality pseudolabels.

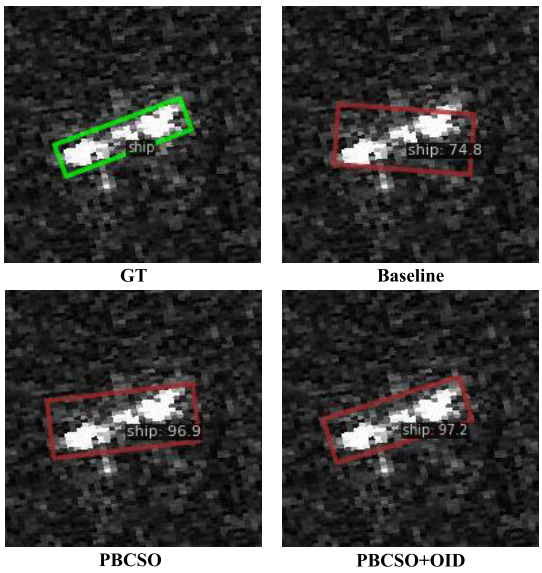


Fig. 7. Predicted OBBs comparisons by applying the PBSCO and OID. The GT are the ground truth of OBB. The 'PBSCO + OID' is the predicted OBB of combination of PBSCO and OID.

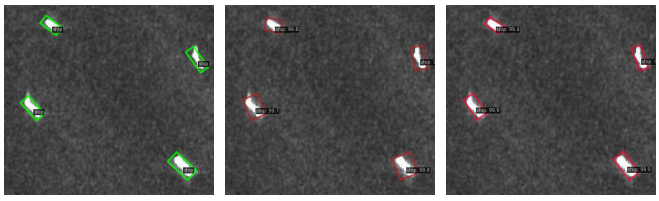


Fig. 8. Predicted OBBs comparisons with/without HL. The GT is the ground truth of OBB. The 'PBSCO + OID' is the predicted OBB of the combination of PBSCO and OID, and the 'PBSCO + OID + HL' is the predicted OBB of the combination of PBSCO, OID, and training by the high-quality pseudolabels.

D. Comparison and Discussion

In these experiments, we compared the performance of our paradigm with popular fully-supervised methods on the SSDD and the HRSID. There are ten fully-supervised methods training by the same configurations, and the results are shown in Tables IV and V. The ten methods can be divided into the single-stage and the multistage, and the R-RetinaNet, the GWD, the R-ATSS, the R-RTMDet, and the S2ANet are the single-stage methods. The multistage methods include the R-Faster-RCNN, the Oriented-RCNN, the R3Det, the ReDet, and the ROI transformer. The first five methods are single-stage methods, and the last five are two-stage methods, in the fully-supervised methods of the Tables IV and V. In addition to the networks specifically designed for the rotated targets detection of remote sensing, other HBB detection networks with prefix 'R-' have been modified for the rotated targets detection. The ATSS assigner in the R-ATSS is using rotated IoU to replace the horizontal IoU of the original assigner in the experiments. The RTMDet with excellent performance is selected as the representative method of the YOLO series. Although its name does not contain 'YOLO', it follows the design style of the YOLO Network.

There are three pairs of AP_{50} and R in Tables IV and V corresponding to entire testset, inshore testset, and offshore testset, respectively, and the inference speed (FPS) and the number of model parameters (Parameters) are in the last two columns of the two tables. The bold values are the optimal

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TABLE V
COMPARISON WITH THE TYPICAL DETECTION METHODS ON HRSID(%)

	Methods	Entire		Inshore		Offshore		FPS(img/s)	Parameters(M)
		R	AP_{50}	R	AP_{50}	R	AP_{50}		
Fully-supervised	R-RetinaNet [6]	79.2	72.7	63.4	48.9	95.0	92.2	19.5	36.13
	GWD [24]	79.1	73.3	64.9	50.4	95.4	93.6	19.1	36.13
	R-ATSS [61]	75.4	71.9	58.0	47.1	96.3	95.7	19.4	36.02
	S2ANet [62]	83.9	80.8	72.2	63.7	95.3	94.6	15.9	36.21
	R-RTMDet [63]	83.9	80.7	71.0	63.3	97.3	96.8	**30.2**	52.26
	R-Faster-RCNN [7]	82.1	77.3	69.7	58.0	96.7	95.0	22.5	41.12
	Oriented-RCNN [27]	84.4	81.7	74.1	67.4	97.4	96.8	21.4	41.13
	R3Det [64]	77.7	73.4	60.4	48.8	95.3	94.6	13.1	41.58
	ReDet [29]	81.9	77.2	68.2	55.7	97.0	95.8	14.3	**31.54**
	RoI Transformer [28]	**86.4**	**83.8**	77.2	70.6	**97.9**	**96.9**	19.8	55.03
Weakly-supervised	Our Paradigm	85.0	81.5	**78.9**	**71.6**	96.2	94.8	16.9	55.84

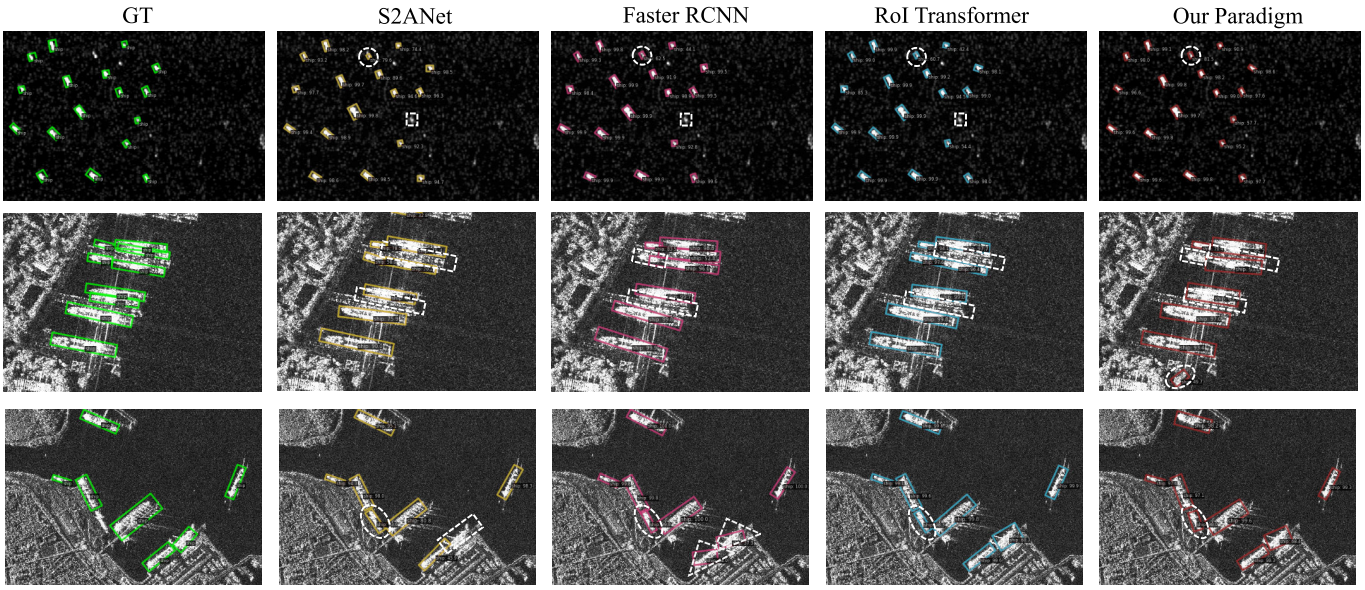


Fig. 9. Visualization results of four representative methods in typical and complex scenarios of the SSDD. The colored rectangles are the detection results. The white and dotted rectangles, circles, and triangles represent the missing ships, the false alarm, and inaccurate OBB, respectively.

values for their columns. Our paradigm obtains AP_{50} of 87.3% and R of 89.7%, AP_{50} of 81.5% and R of 85.0%, in the entire testset of SSDD and HRSID, respectively. Compared with the basal network structure Faster RCNN, this performance is similar to it on SSDD, and gains an improvement of 4.2% and 2.9% on AP_{50} and R on HRSID. The RoI Transformer obtains the best performance on both datasets, with the AP_{50} of 89.5% and R of 91.0% on SSDD, AP_{50} of 83.8% and R of 86.4% on HRSID. In addition, the RoI Transformer can be considered as the fully supervised paradigm corresponding to our WSL paradigm, because the architecture of our WSL paradigm is much closer to the RoI Transformer after removing our proposed components. Due to the lack of supervision information on angle, length, and width, there is a slight gap between fully supervised paradigm and WSL paradigm. Fortunately, this gap is small and our WSL paradigm has the advantage of the inshore test data of HRSID. The WSL paradigm shows the research potential, although this WSL paradigm has just been proposed in SAR ship detection. Our paradigm outperforms the GWD, R3Det, and ReDet on the SSDD as shown in Table IV, and the mainstream OBB detection remote sensing methods on the HRSID except the Oriented RCNN the RoI Transformer, as shown in Table V. On the offshore testset, our paradigm can obtain suboptimal performance on the SSDD, and there is only a gap of AP_{50} of 2.1% and R of 1.7% compared with the best performance on HRSID. In addition, our paradigm gains the best performance in the inshore testset of the HRSID. Although there is a larger percentage gap of inshore testset between our paradigm and the optimal method on SSDD, the actual amount is few due to the small size of the SSDD testset. Therefore, we can draw a conclusion based on these data results, which is our WSL paradigm reaching the performance level of fully-supervised OBB detection methods in remote sensing.

In order to show the performance of our paradigm more comprehensively, the comparison results of three selected representative methods and our paradigm are visualized. They are the S2ANet, the Faster RCNN, and the RoI Transformer, because the S2ANet and the ROI transformer gain the best performance in their types as shown in Tables IV and V, and the Faster-RCNN is the basal structure of our paradigm. Besides, the RoI Transformer can be considered as the fully

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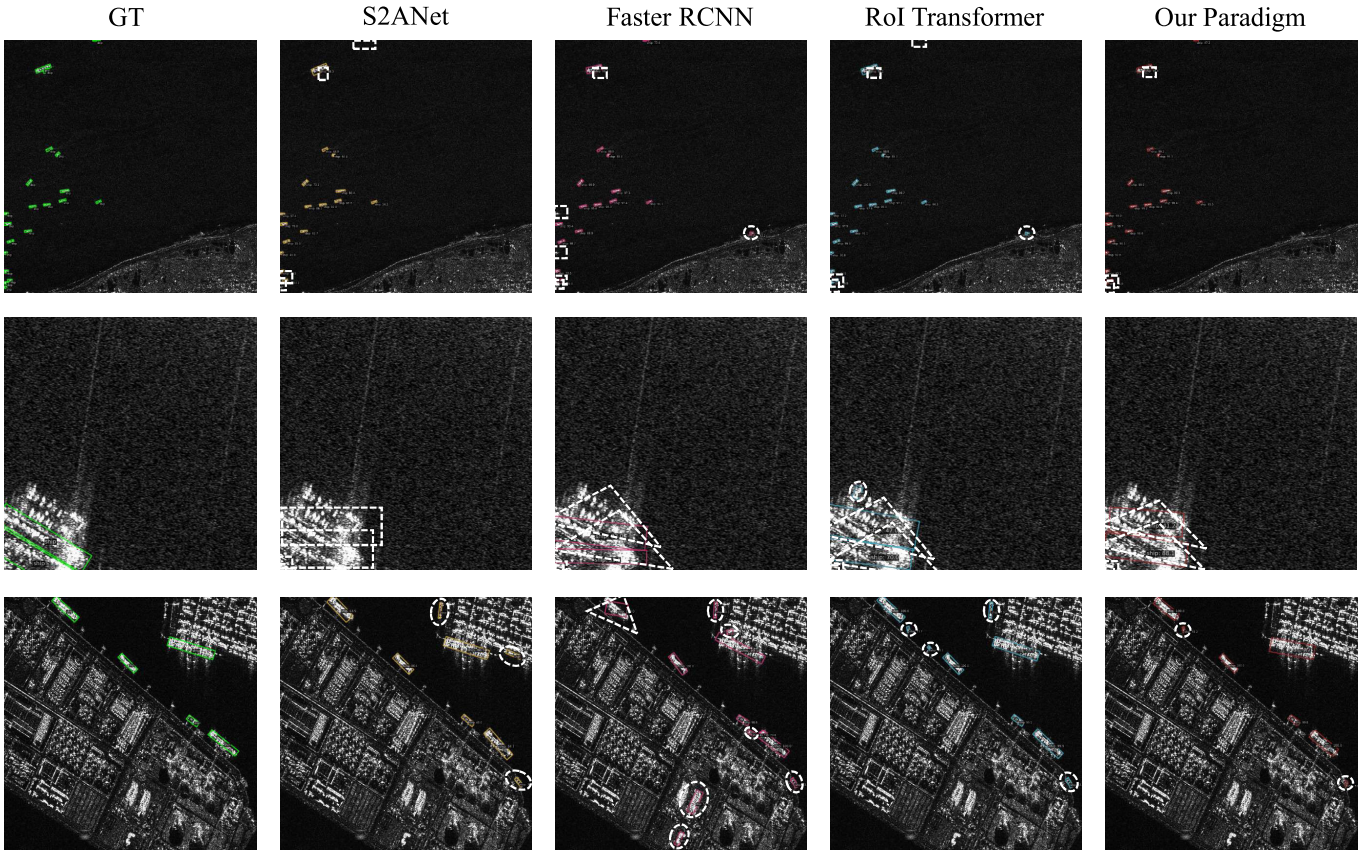


Fig. 10. Visualization results of four representative methods in typical and complex scenarios of the HRSID. The colored rectangles are the detection results. The white and dotted rectangles, circles, and triangles represent the missing ships, the false alarm, and inaccurate OBB, respectively.

supervised paradigm corresponding to our WSL paradigm. The visualization comparisons of the four methods are shown in Figs. 9 and 10, and there are three typical and complex detection scenarios, such as densely docked small ships adjacent to the small island, tightly docked ships and multiscale ships adjacent to the shore. The missing ships and false alarms often appear in the first scenario, and our WSL paradigm outperforms the other three selected fully supervised methods as shown in the first row of both Figs. 9 and 10. The tightly docked ships are easy to detect as a whole ship, which leads to missing ships or inaccurate prediction OBB. The ability to capture ships of our WSL paradigm is as near as dammit with the three methods as shown in the second row of both Figs. 9 and 10, and the detection of tightly docked ships is a troublesome task for the fully-supervised methods. The third scenario is also a troublesome task like the second one, and there are false alarms caused by inshore objects and inaccurate prediction OBB caused by the connection to the coast. There are accurate OBBs for ships, and the false alarm rate is close to the other three methods in our proposed WSL paradigm, as shown in the third row of both Figs. 9 and 10. In a word, the visualization results can also verify our conclusion.

V. Conclusion

To overcome the limitation of fully-supervised OBB detection methods caused by the insufficient OBB annotation data, we proposed a WSL paradigm which follows a coarse-to-fine prediction route in this article, and it is specifically used to train oriented ship detection network in SAR image with only HBB annotation data. The ship orientations predicted by the PBSCO are fed into OID for decoding the OBBs in the coarse prediction stage. These coarse OBBs are the bases of predicting the final OBBs in the fine prediction stage, and the fine prediction head is trained by the combination of high-quality pseudolabels and the HBB annotation. In the training stage of our proposed WSL paradigm, the obstacle of wrong prediction angle and shape both caused by using projection constraint only, is overcome by the coarse ship orientations and high-quality pseudolabels both mined from the SAR image. We verified the performance of the proposed paradigm on the SSDD and HRSID, because they contain both HBB and OBB annotations. The results show that our proposed WSL paradigm can train the network to reach the performance level of the mainstream fully-supervised methods in remote sensing.

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